

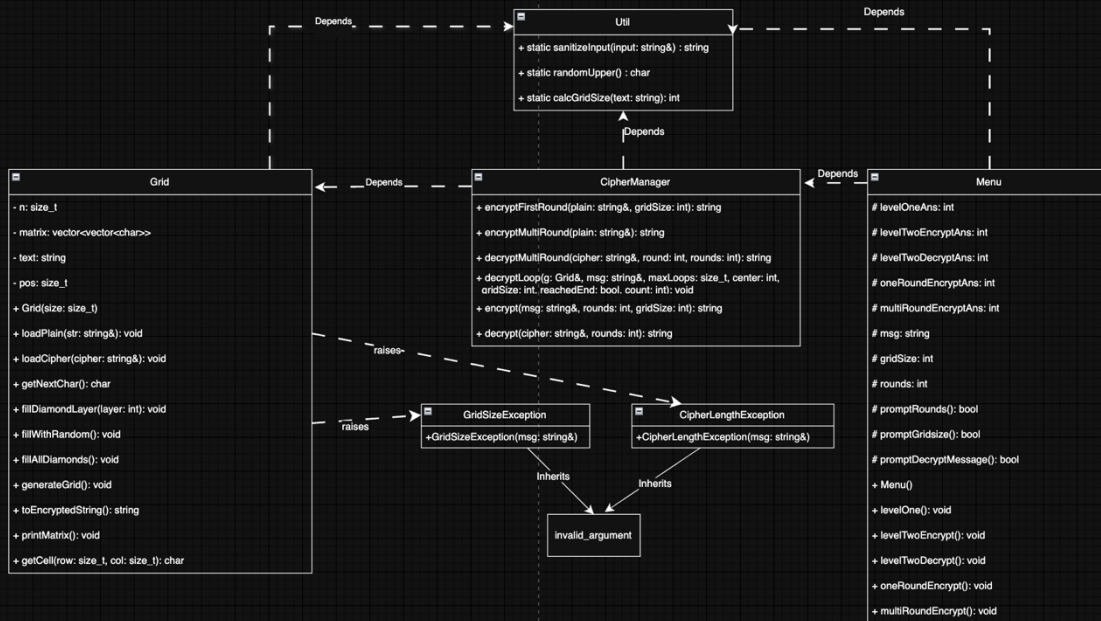
Milestone 1

2800ICT Object Oriented Programming

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Section 1: UML



Section 2: Normal Execution

Scenario 1: Perform one-round encryption on your full name using a manually entered valid grid size. Then, decrypt the corresponding encrypted message. It should include screenshots demonstrating the entire process.

```
(base) jackmillington@Jacks-MacBook-Pro milestone1 % ./week11r
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1

Enter your message: Jack Robert Millington

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1

Enter Gridsize: 7

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
3

Encryption Round 1

M V F K O G R
P X C I R A P
B A L P N O M
J L . V Y G B
C I N S T E Y
N L M O R P F
F Q L T W U A

Message after 1 round: MPBJCNFVXALILQFCL.NMLKIPVSOTORNYTRWGA0GEPURPMBYFA

Encrypted Text:
MPBJCNFVXALILQFCL.NMLKIPVSOTORNYTRWGA0GEPURPMBYFA
```

```

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
4

*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
2

*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
1

Enter your message: MPBJCNFVXALILQFCL.NMLKIPVSOTORNYTRWGAOGEPURPMBYFA

*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
2

Enter Amount of Rounds: 1

*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
3

Decryption Round 1
M V F K O G R
P X C I R A P
B A L P N O M
J L . V Y G B
C I N S T E Y
N L M O R P F
F Q L T W U A

Message after 1 round: JACKROBERTMILLINGTON

JACKROBERTMILLINGTON
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
3

Goodbye!

```

o (base) jackmillington@Jacks-MacBook-Pro milestone1 %

Scenario 2:

Perform one-round encryption on your full name using an automatic grid size. Then, decrypt the corresponding encrypted message. It should include screenshots demonstrating the entire process.

```
● (base) jackmillington@Jacks-MacBook-Pro milestone1 % ./week11r
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1

Enter your message: Jack Robert Millington

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
2

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
3

Encryption Round 1

R X A K Z D Y
G F C I R C E
L A L N N O I
J L . M S G B
K I N M T E X
S Q M O R F O
L U X T U O C

Message after 1 round: RGLJKSLXFALIQUACL.NMXXINMMOTZRNSTRUDCOGEFOYEIBXOC

Encrypted Text:
RGLJKSLXFALIQUACL.NMXXINMMOTZRNSTRUDCOGEFOYEIBXOC
```

```

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
4

*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
2

*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
1

Enter your message: RGLJKSLXFALIQUACL.NMXXKINMMOTZRNSTRUDCOGEFOYEIBXOC

*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
2

Enter Amount of Rounds: 1

*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
3

Decryption Round 1
R X A K Z D Y
G F C I R C E
L A L N N O I
J L . M S G B
K I N M T E X
S Q M O R F O
L U X T U O C

Message after 1 round: JACKROBERTMILLINGTON
JACKROBERTMILLINGTON
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
3

Goodbye!

o (base) jackmillington@Jacks-MacBook-Pro milestone1 %

```

Scenario 3:

Perform automatic multi-round encryption with three rounds on the message below. Then, decrypt the final encrypted output. It should include screenshots demonstrating the entire process.

I ENJOY THIS COURSE.

```
(base) jackmillington@Jacks-MacBook-Pro milestone1 % ./week11r
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1

Enter your message: I ENJOY THIS COURSE.

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
3

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
1

Enter Amount of Rounds: 3

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
2

Encryption Round 1

U B R J C X P
O R N S O L M
S E R C E Y W
I U S Z C . T
K O U Z E H L
M C C G I W J
I N N S T B B

Message after 1 round: UOSIKMIBREU0CNRNRSUCNJSCZZGSC0ECEITXLY.HWBPMWTLJB
```

Encryption Round 2 text: UOSIKMIBREU0CNRNRSUCNJSCZZGSCOECETXLY.HWBPMWTLJB.

K F C T N M M R F E T
G I D D K Z I I R P D
X L K I C H Z B V O F
P U S S . L W G R H O
V O J Y . F H B S E Y
U N L B D C W P P C U
A C X J K I D M O O F
X E U T L H W E C A A
W F L S I T C N W N A
D I L S R E R Z A W Y
S H L N V N A S B O V

Message after 2 rounds: KGXPVUAXWDSFILUONCEFIHCDKSJLXULLTDISYBJTSSNNKC..DKLIRVMZHLFCIHTENMIZWHWDWCRARIBGBPMENZSFRV
POCWABEPOHECOANWOTDFOYUFAAYV

Encryption Round 3 text: KGXPVUAXWDSFILUONCEFIHCDKSJLXULLTDISYBJTSSNNKC..DKLIRVMZHLFCIHTENMIZWHWDWCRARIBGBPMENZSFRV
SPOCWABEPOHECOANWOTDFOYUFAAYV.

Z C T S A C Z A W V A K W H X G H
C U M O W D Q X J D C G D V G Z J
J W D R A K A B M T S E A H S Y Z
H K T F Y U Y N R I S F G W C W I
A Z J D V S E F W V Z S I D B O L
R J H P I T S N X O R W N L R P U
N F X D H Z A Q N W T S H N U L D
E G T I N O . C S P T D P W K O R
K L C E C V J A S E F D F O D C N
W L F M E Y P I X M Q O C W . C O
O H L L P H A W W N Y W C . E Z P
B S Z U H B O A Q U A R D F D G Q
U S G R X Z G P F B A K I N B O T
K I H Z A L M B E R L H W G M Z U
A H B X I P J V I I C B E O R F F
C F J Q T L J S R D O Y C C B W H
A N W R Q G X A K K C N L S M C D

Message after 3 rounds: ZCJHARNEKWOBUKACACUWKZJFGLHSSIHFNMTDTJHXT CFLZGHBJSORFDPDIEMLURZXQRAWAYVIHNCEPHXAITQCDKUSTZ
VYHBZLPLGZQAYESA. JPAOGMJJXAXBNFNQCAIWAPBVSJMRWXNSSXWQFEIRKVDITIVOWPEMNUBRIDKACSSZRTTFQYAALCOCKGEFSWSDDOWRKHBYNWDAGI
HPFCCDIWECLHVHWDLNWOW. FNGOCSXGSCBRUKD. EDBMRBMGZYWOPLCCZGOZFWCHJZILUDRNOPQTUFHD

Encrypted Text:

ZCJHARNEKWOBUKACACUWKZJFGLHSSIHFNMTDTJHXT CFLZGHBJSORFDPDIEMLURZXQRAWAYVIHNCEPHXAITQCDKUSTZOVYHBZLPLGZQAYESA. JPAOGM
JXAXBNFNQCAIWAPBVSJMRWXNSSXWQFEIRKVDITIVOWPEMNUBRIDKACSSZRTTFQYAALCOCKGEFSWSDDOWRKHBYNWDAGINHPFCCDIWECLHVHWDLNWOW. F
NGOCSXGSCBRUKD. EDBMRBMGZYWOPLCCZGOZFWCHJZILUDRNOPQTUFHD

* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *

4

* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *

2


```
*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
1
```

Enter your message: ZCJHARNEKw0BUKACACUwKZJFGLHSSIHfNTMDTJHXTcFLZGHBjWSORFDPDIeMLURZXQRAWAYVIHnCEPHXAITQCDKUSTZ0VYHB
ZLPLGZQAYESA.JPA0GMJJXAXBNFNQCAIWAPBVSawJMRwXNSXWQFEIRKVDITIVOWPeMNUBRIDKACSSZRTTFQYAALCOCKGEFSWDDOWRKHBYNWDAGINHPFC
CDIWECLHvHvDLNwOW.FNG0CSXGSCBRUKD.EDBMRBMGZYwOPL0CCZG0ZFWCHJZILUDRN0PQTUFHD

```
*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
2
```

Enter Amount of Rounds: 3

```
*****
* Menu - Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
3
```

Decryption Round 1

Z	C	T	S	A	C	Z	A	W	V	A	K	W	H	X	G	H
C	U	M	O	W	D	Q	X	J	D	C	G	D	V	G	Z	J
J	W	D	R	A	K	A	B	M	T	S	E	A	H	S	Y	Z
H	K	T	F	Y	U	Y	N	R	I	S	F	G	W	C	W	I
A	Z	J	D	V	S	E	F	W	V	Z	S	I	D	B	O	L
R	J	H	P	I	T	S	N	X	O	R	W	N	L	R	P	U
N	F	X	D	H	Z	A	Q	N	W	T	S	H	N	U	L	D
E	G	T	I	N	O	.	C	S	P	T	D	P	W	K	O	R
K	L	C	E	C	V	J	A	S	E	F	D	F	O	D	C	N
W	L	F	M	E	Y	P	I	X	M	Q	O	C	W	.	C	O
O	H	L	L	P	H	A	W	W	N	Y	W	C	.	E	Z	P
B	S	Z	U	H	B	O	A	Q	U	A	R	D	F	D	G	Q
U	S	G	R	X	Z	G	P	F	B	A	K	I	N	B	O	T
K	I	H	Z	A	L	M	B	E	R	L	H	W	G	M	Z	U
A	H	B	X	I	P	J	V	I	I	C	B	E	O	R	F	F
C	F	J	Q	T	L	J	S	R	D	O	Y	C	C	B	W	H
A	N	W	R	Q	G	X	A	K	K	C	N	L	S	M	C	D

Message after 1 rounds: KGXPVUAXWDSFILUONCEFIHCDSJLXULLLTDISYBJTSSNNKC..DKLIRVMZHLFCIHTEHNIMIZWHWDWCRARIBGBPMENZSFRVRS
POCWABEPOHECOANWOTDFOYUFAAYV

```
Decryption Round 2
K F C T N M M R F E T
G I D D K Z I I R P D
X L K I C H Z B V O F
P U S S . L W G R H O
V O J Y . F H B S E Y
U N L B D C W P P C U
A C X J K I D M O O F
X E U T L H W E C A A
W F L S I T C N W N A
D I L S R E R Z A W Y
S H L N V N A S B O V
```

Message after 2 rounds: UOSIKMIBREUOCNRRNSUCNJSCZZGSCOECEITXLY.HWBPMWTLJB

```
Decryption Round 3
U B R J C X P
O R N S O L M
S E R C E Y W
I U S Z C . T
K O U Z E H L
M C C G I W J
I N N S T B B
```

Message after 3 rounds: IENJOYTHISCOURSE

```
IENJOYTHISCOURSE
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
3
```

Goodbye!

Section 3: Abnormal Execution

Level 1 Menu:

Non Numeric Text:

```
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
one

Please enter only 1, 2, or 3
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
```

Out of range numbers:

```
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
5

Please enter only 1, 2, or 3
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
```

Floats:

```
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1.2

Please enter only 1, 2, or 3
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
```

Mixed:

```
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
2a

Please enter only 1, 2, or 3
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
```

Empty (just clicked enter):

```
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****

2

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
```

Whitespace Only:

```
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
```

Both:

Nothing will happen until some input is entered, Same with a whitespace only entry:

```
Please enter only 1, 2, or 3
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
de#%^421

Please enter only 1, 2, or 3
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
```

Level 2 – Encryption Menu:

Empty/ whitespace / non-numeric text:

```
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****

ede

Please enter only 1, 2, 3 or 4
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
```

Out of range Number:

```
Please enter only 1, 2, 3 or 4
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
53

Please enter only 1, 2, 3 or 4
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
```

Decimal/mixed

```
Please enter only 1, 2, 3 or 4
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2.3f

Please enter only 1, 2, 3 or 4
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
```

Level 2 Encryption menu – Option 1, enter a message

Whitespace only message:

```
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1
Enter your message:
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
2
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
3
Encryption Round 1
.
Encrypted Text:
.
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
#
```

Punctuation Only:

```
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1
Enter your message: !!!...
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
2
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
3
Encryption Round 1
P P ! O I
V ! N . A
! C C V .
R V M . X
G O J C P
Encrypted Text:
PV!RG6P!CQ0!NCM10-V,CTA,XP
```

Emojis and non ASCII:

```
Enter your message: 🐼
Error: message cannot have non ASCII Characters.
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
|
```

Menu - Level 3, One round Encryption:

Menu - Level 3, One round Encryption –

It will not take anything other than 1,2,3 or 4 as seen here:

Enter gridSize (string, emoji, over 99, even, empty):

```
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
a
Please enter only 1, 2, 3 or 4
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
####@$$
Please enter only 1, 2, 3 or 4
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
.
Please enter only 1, 2, 3 or 4
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
🐼
Please enter only 1, 2, 3 or 4
```

```
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1
Enter Gridsize: fwe
Error: please enter an odd integer.
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1
Enter Gridsize: 🐼
Error: please enter an odd integer.
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1
Enter Gridsize: 333
Error: gridSize must be an odd integer between 1 and 99
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1
Enter Gridsize: 32
Error: gridSize must be an odd integer between 1 and 99
*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1
Enter Gridsize:
Error: gridSize cannot be empty.
```

Menu – Level 3 – Multi Round Encryption:

String, emoji, out of range number,
float, punctuation, whitespace:

```
*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
edf

Please enter only 1, 2, or 3
*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
👉👉👉

Please enter only 1, 2, or 3
*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
4

Please enter only 1, 2, or 3
*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
1.2

Please enter only 1, 2, or 3
*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
!.d

Please enter only 1, 2, or 3
*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****

f

Please enter only 1, 2, or 3
```

Option 1 – Enter Round Number

Out of range number, string, empty:

```
Enter Amount of Rounds: 14
Error: rounds must be within 1 to 10.

*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
1

Enter Amount of Rounds: 0
Error: rounds must be within 1 to 10.

*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
1

Enter Amount of Rounds: one
Error: please enter a valid integer.

*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
1

Enter Amount of Rounds:
Error: round count cannot be empty.

*****
* Menu – Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and *
* the corresponding encoded message *
* 3. Back *
*****
```

Menu – Level 2 Decryption:

Out of range, string, non-ASCII
whitespace + multi click enter

```
*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
0

Please enter only 1, 2, 3 or 4
*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
fefefef

Please enter only 1, 2, 3 or 4
*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
ede

Please enter only 1, 2, 3 or 4
*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
😄

Please enter only 1, 2, 3 or 4
*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
█
```

Option 2: Enter the round number

Out of range input, string, empty.

```
Enter Amount of Rounds: 0
Error: rounds must be within 1 to 10.

*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
2

Enter Amount of Rounds: 14
Error: rounds must be within 1 to 10.

*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
2

Enter Amount of Rounds: one
Error: please enter a valid integer.

*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
2

Enter Amount of Rounds:
Error: round count cannot be empty.

*****
* Menu – Level 2: Decryption *
* Select an option: *
* 1. Enter a message *
* 2. Enter the round number *
* 3. For each round, print the grid and *
* the corresponding decoded message *
* 4. Back *
*****
█
```

Option 1: Enter a Message:

Cipher is not $n \times n$ length:

```
*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
Enter your message: dedededddd

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
Enter Amount of Rounds: 1

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
3

Decryption Round 0
Error: Cipher must be of  $n \times n$  length where  $n$  is odd

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****

```

Non ASCII at correct length:

```
Enter your message: 🍌🍌🍌🍌🍌🍌🍌.

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
2

Enter Amount of Rounds: 1

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
3

Decryption Round 0
Error: Cipher must be of  $n \times n$  length where  $n$  is odd

```

Whitespace and
enter clicked before
valid input:

```
Enter your message:

          nineleter

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
2

Enter Amount of Rounds: 1

*****
* Menu - Level 2: Decryption *
* Select an option:         *
* 1. Enter a message        *
* 2. Enter the round number *
* 3. For each round, print the grid and *
*   the corresponding decoded message *
* 4. Back                   *
*****
3

Decryption Round 0
n e t
i l e
n e r

ieeel\nntr
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****

```

Unique:

Not specifying a grid size on single round encryption: No Round Selected:

```
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1

Enter your message: hello

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
3

Error: Please include a gridSize.
```

```
(base) jackmillington@Jacks-MacBook-Pro milestone1 % clang++ -std=c++17 main.cpp -O2 -o m
(base) jackmillington@Jacks-MacBook-Pro milestone1 % ./m
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1

Enter your message: hello

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
3

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and the corresponding encoded message *
* 3. Back *
*****
2

Error: no round count specified, Please choose option 1 first.

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter the round number *
* 2. For each round, print the grid and the corresponding encoded message *
* 3. Back *
*****
```

Grid size is too small to fit message:

```
*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
1

Enter your message: hellooo

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1

Enter Gridsize: 3
Error: gridsize is too small to fit message.

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
```

No message is ever entered:

```
(base) jackmillington@Jacks-MacBook-Pro milestone1 % clang++ -std=c++17 main.cpp -O2 -o m
(base) jackmillington@Jacks-MacBook-Pro milestone1 % ./m
*****
* Menu - Level 1 *
* Select an option: *
* 1. Encrypt a message *
* 2. Decrypt a message *
* 3. Quit *
*****
1

*****
* Menu - Level 2: Encryption *
* Select an option: *
* 1. Enter a message *
* 2. One-round encryption *
* 3. Automatic multi-round encryption *
* 4. Back *
*****
2

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
1

Enter Gridsize: 3

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
3

Error: no message entered. Please go back and enter a message.

*****
* Menu - Level 3: Encryption *
* Select an option: *
* 1. Enter a grid size *
* 2. Automatic grid size *
* 3. Print the grid and the encoded message *
* 4. Back *
*****
```


Section 4: Application of course concepts/ knowledge from week 1 to week 7

Week 1:

- Using namespace std
- c++ auto
- basic input / output
- variable definitions
- control-flow
- if else statements
- for and while loops
- switch statements
- fundamental data types
- arithmetic operators
- logical operators
- string class and methods
- vectors and methods
- functions
- reference variables
- C++ Include Files
- exit()
- numeric_limits<streamsize>::max()

Week 2:

- Iterators
- Auto iterators
- .begin()
- .end()
- [] operator
- Vector class constructors
- Initialising vectors
- Adding new elements to vectors
- Iterators with vectors
- Vector common member functions
- Retrieving elements

Week 3:

- STL Algorithms
- Std::all_of()

Week 4:

- C++ Classes
- Access Specifiers
- Private
- Public
- Using const with member functions
- Inline member functions
- Constructors

Week 5:

- Static member variable
- UML Class Diagram
- UML Access Specification Notation
- UML Data Type Notation
- UML Parameter Type Notation
- UML Function Return Type Notation

Week 6:

- Calling Classes from Inside another class
- Class inheritance
- Protected members
- Constructors

Week 7:

- Exceptions
- Throw
- Try
- Catch
- Handling Multiple Exceptions
- e.what()
- <stdexcept>
- Std::getline()
- Lambda Expressions
- Anonymous Function Objects